

**Objective:** I can solve multi-step cube root equations, equations with coefficients on both sides, and rational exponent equations.

## Quick Review: The Strategy

1. **ISOLATE:** Get the radical ( $\sqrt[3]{x}$ ) or the rational exponent term ( $(x)^{\frac{2}{3}}$ ) by itself.
2. **ELIMINATE:** Raise both sides to the **reciprocal power** to remove the exponent.
3. **SOLVE:** Finish using standard algebra.

## Part 1: Coefficients on Both Sides (The "Trap")

When you have radicals on both sides with numbers in front, you must cube the **ENTIRE** side.

### Key Concept: Distributing the Power

When you cube a product, the power goes to **both** the number and the radical:

$$(a\sqrt[3]{b})^3 = a^3 \cdot b$$

### Example A: Coefficients

$$4\sqrt[3]{x} = 2\sqrt[3]{x+7}$$

#### Step 1: Cube BOTH sides

$$(4\sqrt[3]{x})^3 = (2\sqrt[3]{x+7})^3$$

#### Step 2: Apply power to coefficients

$$4^3 \cdot x = 2^3 \cdot (x+7)$$

$$64x = 8(x+7)$$

#### Step 3: Solve

$$64x = 8x + 56 \implies 56x = 56 \implies x = 1$$

**Watch Out!** A common mistake is forgetting to cube the 4 and the 2.

## Part 2: Rational Exponents (Reciprocal Powers)

When the exponent is a fraction like  $\frac{2}{3}$ , we raise both sides to the **reciprocal** (flip the fraction).

### Example B: Practice Problems

#### 1. Numerator is not 1

$$x^{\frac{2}{3}} = 49$$

#### 2. Coefficient & Reciprocal

$$2x^{\frac{3}{2}} = 54$$

### Even Numerator Alert!

If the **numerator** of the original exponent is **EVEN** (like  $\frac{2}{3}$  or  $\frac{4}{5}$ ), you are effectively taking an even root. This means you need  $\pm$  in your answer (just like  $x^2 = 25 \rightarrow x = \pm 5$ ).

## Part 3: Multi-Step Practice

*Isolate first, then use the reciprocal power.*

#### 1. Isolate First:

$$2(x)^{\frac{2}{3}} + 5 = 55$$

#### 2. Grouped Expression:

$$(x - 2)^{\frac{4}{3}} = 16$$

## Part 4: Application

### Biology Application

The weight of a mammal  $W$  (in kg) can be estimated by the formula  $W = 0.5C^{\frac{3}{2}}$ , where  $C$  is the circumference of its chest (in meters).

**Problem:** If a mammal weighs 32 kg, what is the circumference of its chest?